Project Title: Design and analyze a Pacemaker Electrode using COMSOL Multiphysics

BME 3114: Simulation and Modeling Laboratory

A project submitted to the Department of Biomedical Engineering in Third Year First Term

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Abstract

This project focuses on the design and analysis of pacemaker electrodes using COMSOL Multiphysics, a powerful finite element analysis software. Pacemaker electrodes play a vital role in regulating cardiac rhythm by delivering precise electrical stimulation to the heart. The project aims to design the pacemaker electrode and analyse the current and voltage distribution for different materials and different radius of the tip of the electrode.

Key objectives include the creation of detailed three-dimensional models of pacemaker electrodes and surrounding tissue, incorporating realistic geometries, material properties, and boundary conditions to avoid shock and current flow out side of the tissue of interest. The simulation platform will allow for analysis of current and voltage distribution throughout the electrode for different materials and also figuring out the suitable material for the electrode, and last but not the least analysis of the electrodes function and current and voltage distribution changing the radius of the tip of the electrode.

Through rigorous analysis and simulation, the project seeks to enhance our understanding of the electrical behavior of pacemaker electrodes and their interaction with cardiac tissue. Insights gained from the simulations will inform the design process, leading to the development of next-generation pacemaker electrodes with enhanced performance and reliability.

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Chapter 1

Introduction

1.1 What is a pacemaker

A pacemaker is a medical device which is used to maintain normal rhythmic activity of the heart. It consists of a pulse generator and one or more leads. The pulse generator is an electronic device which is capable of generating electric pulses for stimulating the heart muscles as given in [1][2].

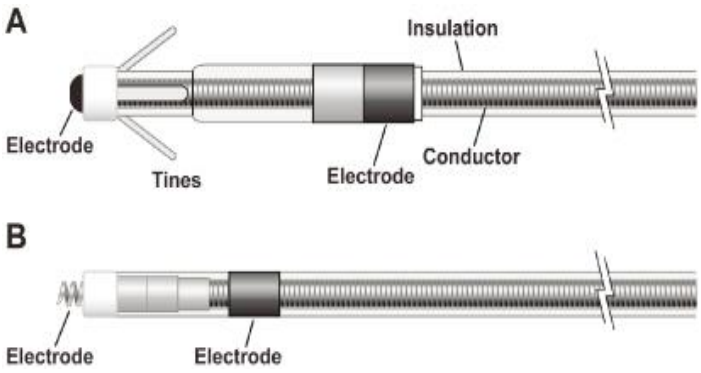


Figure 1: Schematic of Passive Fixation (A) and Active Fixation (B)

Figure 1 shows the schematic drawing of two types of fixation wires used in Heart Pacemaker. In passive fixation, the term “passive” indicates only that the lead is attached to the heart muscle without a mechanism that holds the end of the lead against the heart muscle, most commonly a small corkscrew like device on the end of the lead. On the other hand, Active-fixation pacing leads allow the use of selective pacing sites [3].

1.2 Why is it used

Arrhythmias are due to cardiac problems producing abnormal heart rhythms. In general arrhythmias reduce hemodynamic performance including situations where the heart's natural pacemaker develops an abnormal rate or rhythm or when normal conduction pathways are interrupted and a different part of the heart takes over control of the rhythm. An arrhythmia can involve an abnormal rhythm increase (tachycardia; > 100 bpm) or decrease (bradycardia; < 60 bpm), or may be characterized by an irregular cardiac rhythm, e.g. due to asynchrony of the cardiac chambers. An "artificial pacemaker" can restore synchrony between the aria and ventricles [4].

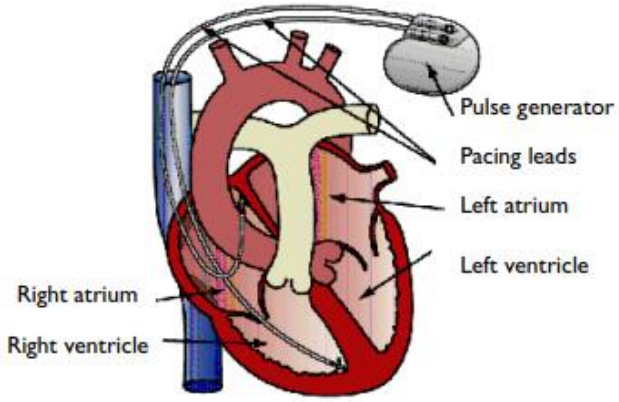


Figure 2: Schematic drawing of the heart with two pairs of pacemaker electrodes

Figure-2 shows a schematic drawing of two pair of electrodes placed inside the heart. The electrodes are supplied with current from the pulse generator unit, which is also implanted in the patient.The modeled device is a pacemaker electrode that is placed inside the heart and helps the patient's heart to keep a normal rhythm. The device is referred to as an electrode, but it actually consists of two electrodes: a cathode and an anode.

1.3 Model definition

The model domain consists of the blood and tissue surrounding the electrode pair. The actual electrodes and the electrode support are boundaries to the modeled domain. The working electrode consists of a hemisphere placed on the tip of the supporting cylindrical structure. The counter electrode is placed in the "waist" of this structure. All other surfaces of the supporting structure are insulated. The outer boundaries are placed far enough from the electrode to give a small impact on the current and potential distribution [5]. In COMSOL Multiphysics, use the 3D Conductive Media DC application mode for the analysis of the electrode. This application mode is useful for modeling conductive materials where a current flows due to an applied electric field.

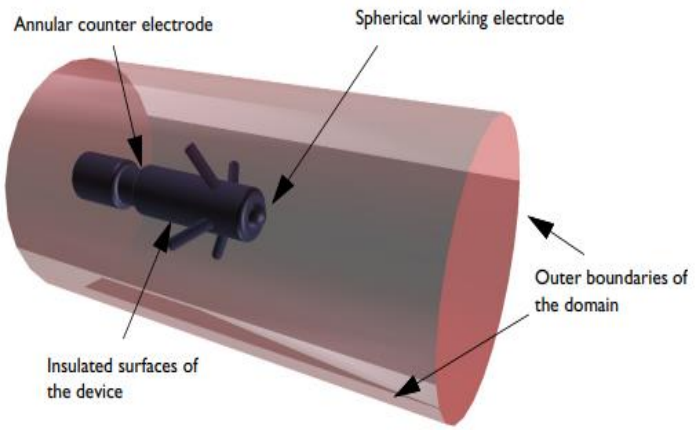


Figure-3: Modeling domain and boundaries

The working electrode consists of a hemisphere placed on the tip of the supporting cylindrical structure. The counter electrode is placed in the “waist” of this structure. All other surfaces of the supporting structure are insulated. The outer boundaries are placed far enough from the electrode to give a small impact on the current and potential distribution.
In COMSOL Multiphysics, use the Electric Currents interface for the analysis of the electrode. This physics interface is useful for modeling conductive materials where a current flows due to an applied electric field.

1.4 Domain equations

The current in the domain is controlled by the continuity equation, which follows from Maxwell’s equations and its also called the Continuity Equation for Electric Current.

$$-\nabla\cdot(\sigma\nabla V)=0$$

where σ is the conductivity of the human tissue and V is the voltage. This equation describes how electric current (J) flows through a conductive medium, such as human tissue. It states that the divergence of the conductivity (σ) times the gradient of the electric potential (V) is equal to zero, indicating that current is conserved, i.e., it neither accumulates nor depletes within the tissue. This equation helps to understand how the electric potential (V) gradient and conductivity (σ) influence the distribution of current around the electrode tip.

This equation uses the following relations between the electric potential and the fields.

$$\mathrm{E} = -\nabla\mathrm{V}$$

The electric field (E) is crucial for understanding how the electric potential (V) changes with respect to spatial coordinates around the pacemaker electrode. In the context of a pacemaker, this equation helps to determine the strength and direction of the electric field generated by the electrode. This electric field is what influences the nearby cardiac tissue, potentially depolarizing cells and causing them to contract, thereby regulating the heart's rhythm.
These equations provide a theoretical framework for understanding the electrical behavior of pacemaker electrodes and their interaction with surrounding tissue. By applying these equations, researchers and clinicians can optimize electrode design and stimulation parameters to ensure effective and safe cardiac pacing therapy.

1.5 Boundary conditions

Ground potential boundary conditions imply that the electrode's thinner waist is connected to a reference point, often considered as "ground." This connection ensures that the potential at the thinner waist of the electrode remains constant and equal to the ground potential. In the context of a pacemaker electrode, this boundary condition helps to stabilize the electrical potential distribution and ensures that the electrode operates within a defined voltage range. The fixed potential of 1 V at the electrode tip means that the electric potential at the tip remains constant and independent of other factors. This fixed potential is typically set by the pacemaker device to ensure consistent stimulation of the surrounding tissue. In practical terms, it means that the pacemaker electrode delivers a controlled voltage to stimulate the cardiac tissue, thereby regulating the heart's rhythm. All other boundaries are electrically insulated.

$$n\ J\cdot = 0$$

The condition $n \cdot J = 0$ signifies that the electric current (J) flowing through the boundary of the pacemaker electrode is perpendicular. (n denotes the outward normal vector to the boundary). In practical terms, this means that there is no net flow of electric current across the boundary.

In the context of a pacemaker electrode, this condition ensures that the electrical stimulation provided by the electrode remains confined within the tissue region of interest, typically the cardiac muscle. It implies that the electrode's stimulation does not leak or spread beyond its intended target area.

Chapter 2

Motivation/ Background Study

2.1 General discussion

The pacemaker is one of the revolutionary inventions in the medical field. It has the ability to treat cardiovascular diseases such as arrhythmias. By its nature, it provides small electric pulses to try and get the heart to beat normally again. The pacemaker electrode is a very important part of the whole pacemaker circuit. The pacemaker electrode is the part that actually attaches with the heart muscle and provide the electric pulse.

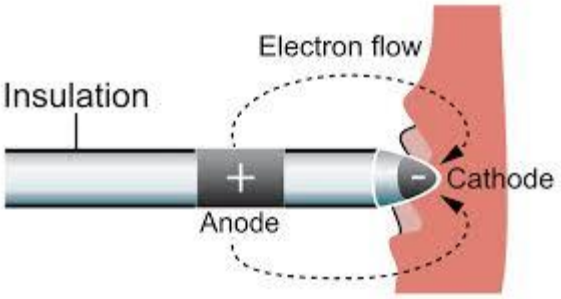


Figure 4: Pacemaker electrode fixation principal [6]

2.2 Main concerns

The potential distribution and the current density from the lead is an issue of concern here. The heart is quite obviously one of the most important and delicate organs. The electric pulses need to confined within a small area of interest. The current must not be too high for the heart to handle. The mean acute current threshold is about 1mA and the chronic current might range from 1 to 5mA. By the design and analysis of the pacemaker electrode, we can predict the potential distribution across the electrode and also visualise the streamlines of the current density. The vital requirements for the electrode are-

Highest current density at the tip using low current electronics

Proper insulation of the electrode body

So, the examination of the pacemaker electrode before attaching it either rest of the circuit is vital. Using COMSOL Multiphysics ® the pacemaker electrode can be both designed and analysed under the appropriate physics.

Chapter 3

Methodology

3.1 Procedure in brief

The procedure adapted in order to design and analyse a pacemaker model using COMSOL Multiphysics ® is given below -

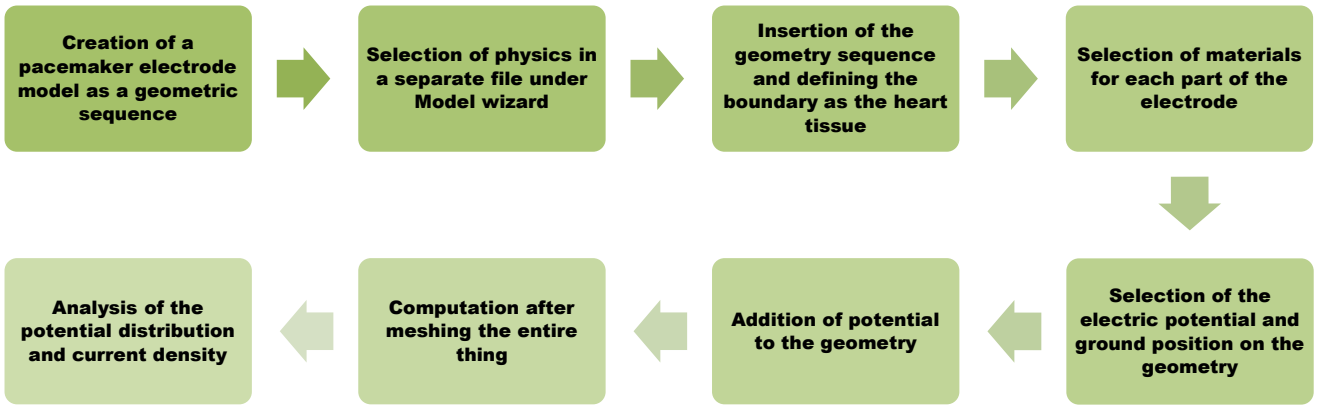


Figure 5: A flowchart depicting the workflow for the task

Chapter 4

Implementation

4.1 Design

1. At first a blank model had been opened for designing the 3D model of the pacemaker electrode. Then from the add component toolbar a 3D component has been added and from the geometry tab a work plane has been added on the yz plane.

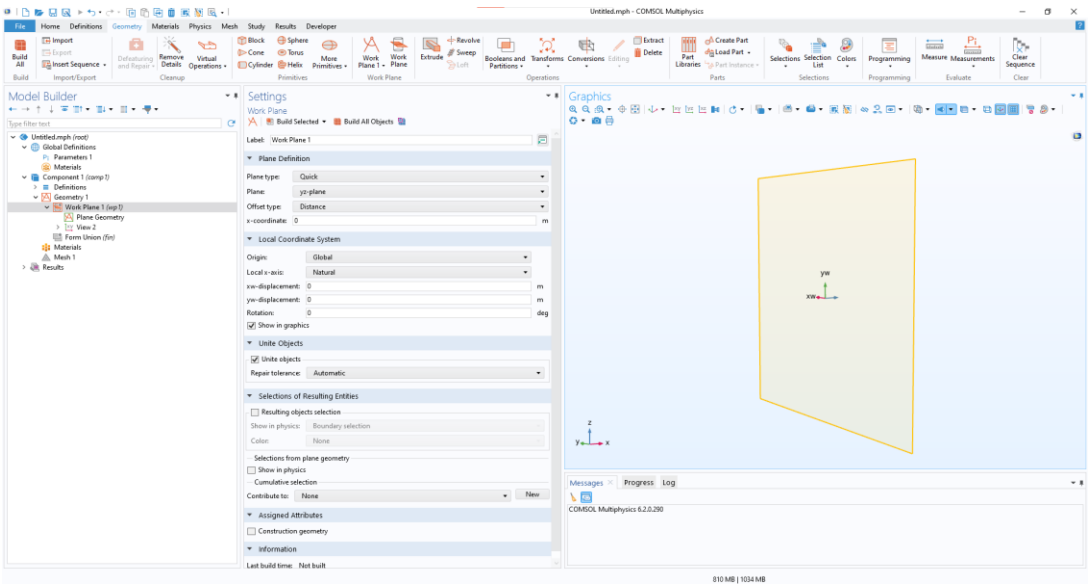


Figure 6: Work plane (YZ plane)

2. Then on the plane geometry a rectangle had been sketched from the center (0,0). The width of the rectangle is rect1_w = 2.1 mm and the height is rect1_h = 20 mm.

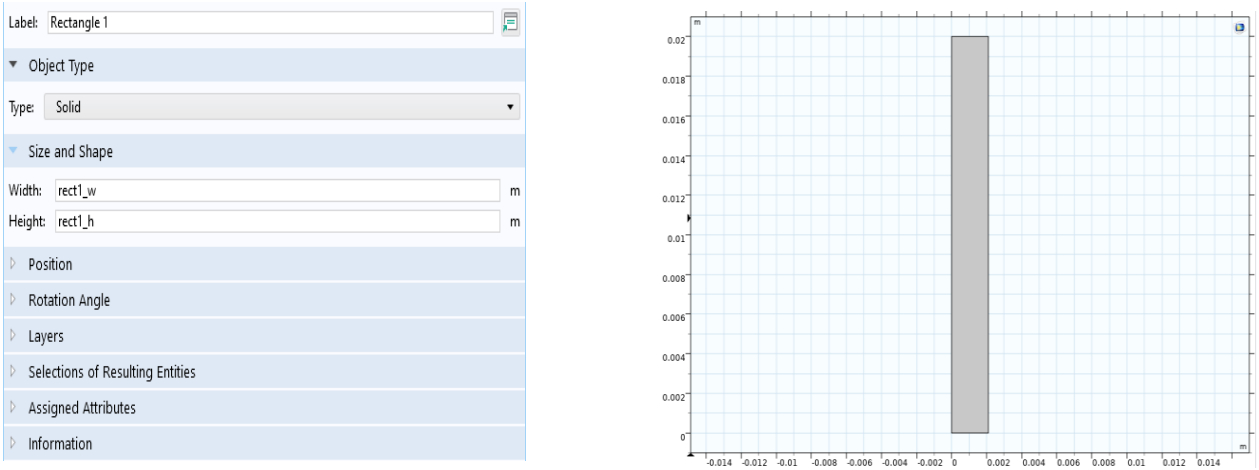


Figure 7: Sketch on the plane geometry

3. Then another rectangle (r2) with width rect2_w = 0.5 mm and height rect2_h = 3 mm had been sketched and subtracted from the 1st rectangle.

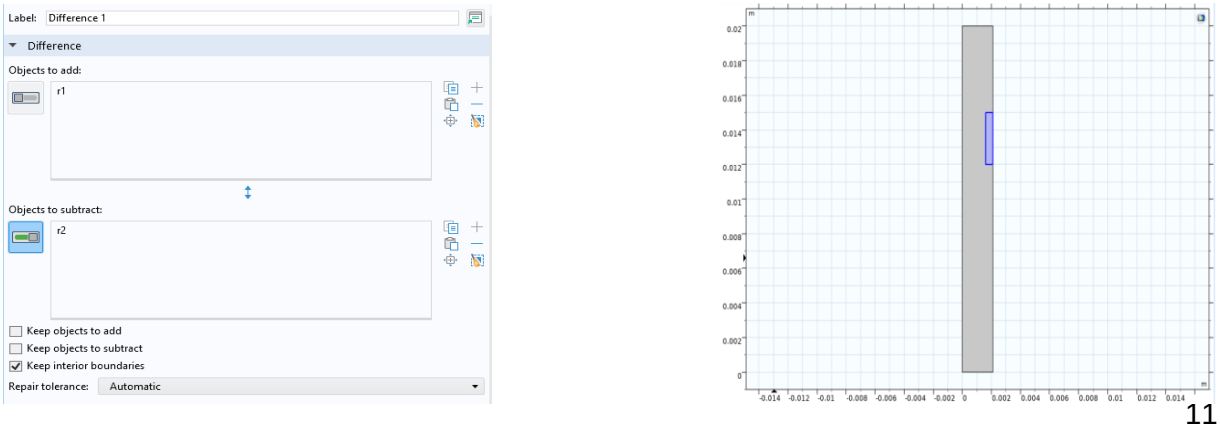


Figure 8: Sketch on the plane geometry to

4. Then fillet had been applied at the corner 5, 6, and 7 with fillet radius fil_rad = 0.5 mm.

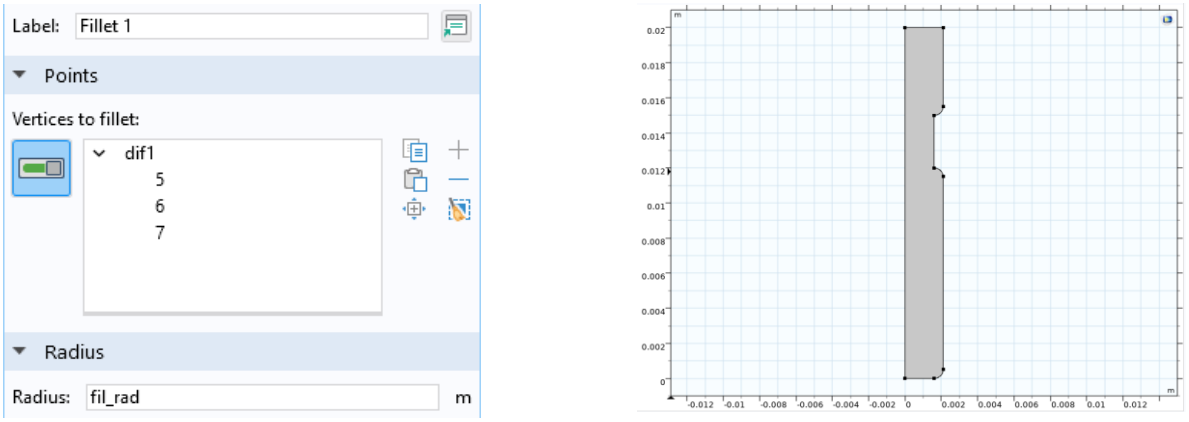


Figure 9: Fillet the sharp corner (5, 6, and 7)

5. After that the 2D sketch has been revolved to make the 3D model.

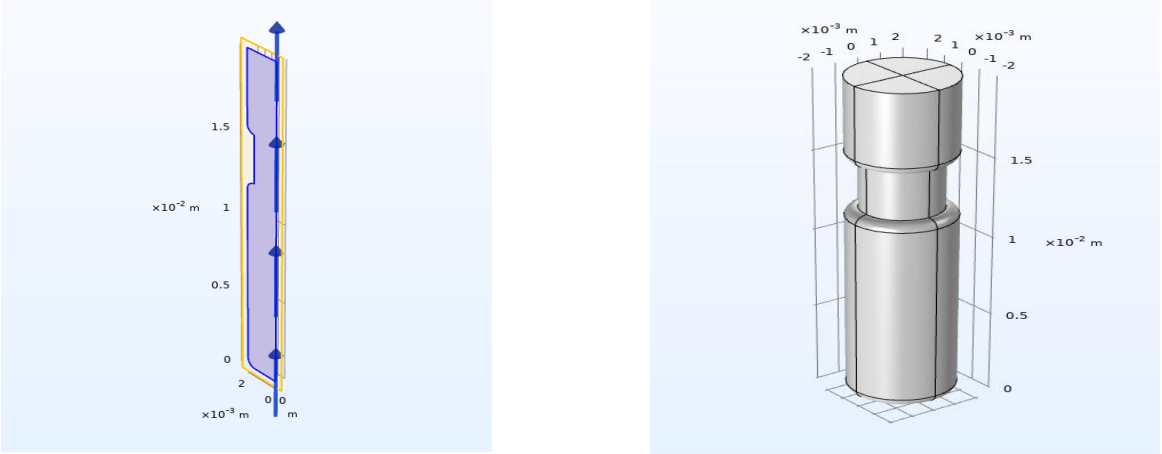


Figure 10: 3D structure after revolving the 2D

6. Then a sphere had been taken at the center (0,0) with radius sph_electrode=1mm. This is the spherical working electrode.

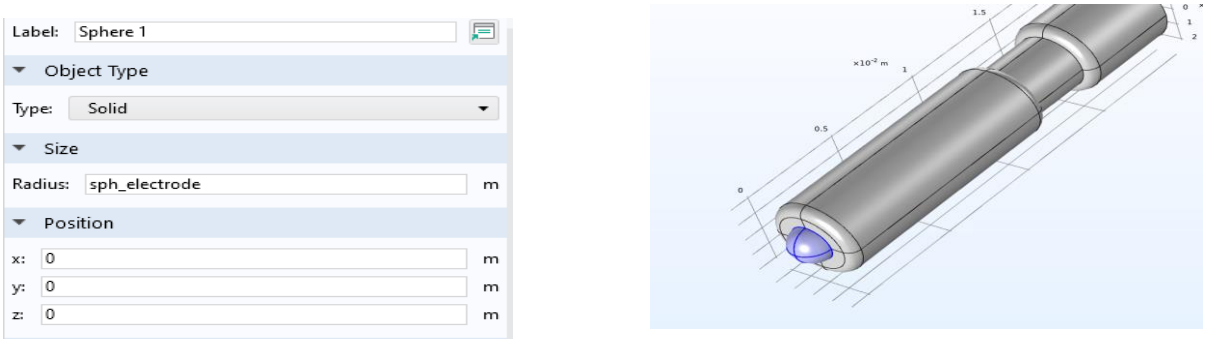


Figure 11: Sphere electrode

7. Then yz plane had been added for the 2nd work plane to sketch the anchoring arm.

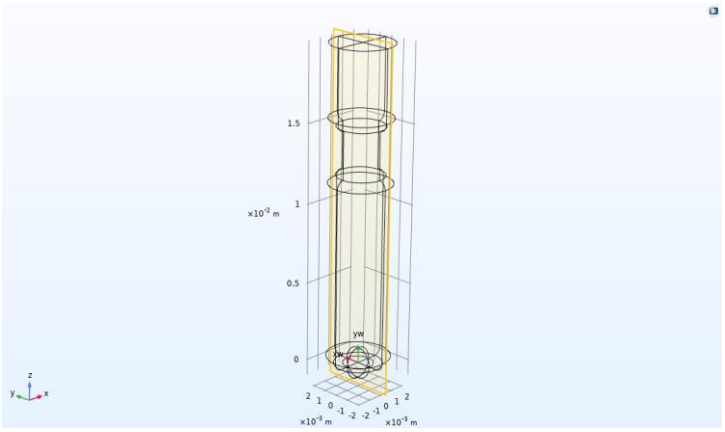


Figure 12: 2nd work plane

8. On the 2nd work plane, a rectangle had been sketched with width wp2_rect1_w= 0.5mm and height wp2_rect1_h = 5 mm. The position of the rectangle was 0.8 mm along xz direction and 2 mm along yz direction from the corner. Then chamfer had been applied to one corner of the rectangle.

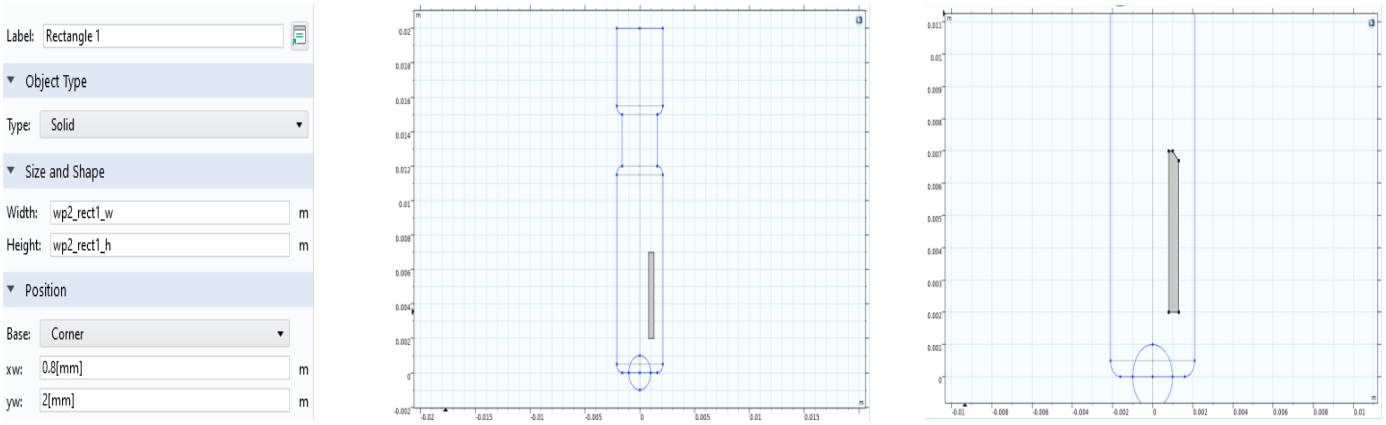


Figure 13: Sketch for anchoring arm (distance from vertex is 0.3 mm for

9. Then the 2D sketch of the anchoring arm had been revolved where the revolution axis was kept 0.8 mm away from the center along xz direction.

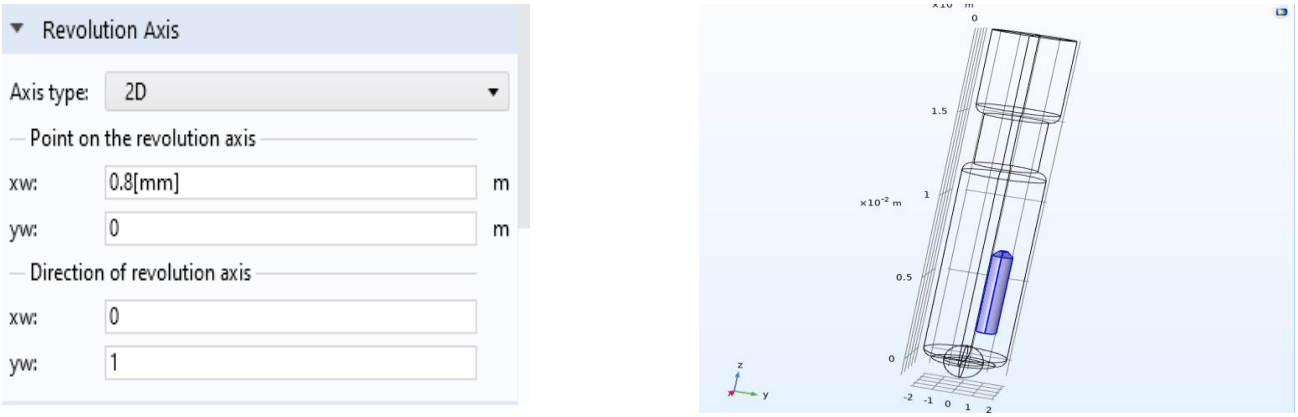


Figure 14: 3D structure after revolving the 2D sketch of anchoring arm

10. Then the anchoring arm had been rotated to the right with an angle of rotation of -60 degree from x axis.

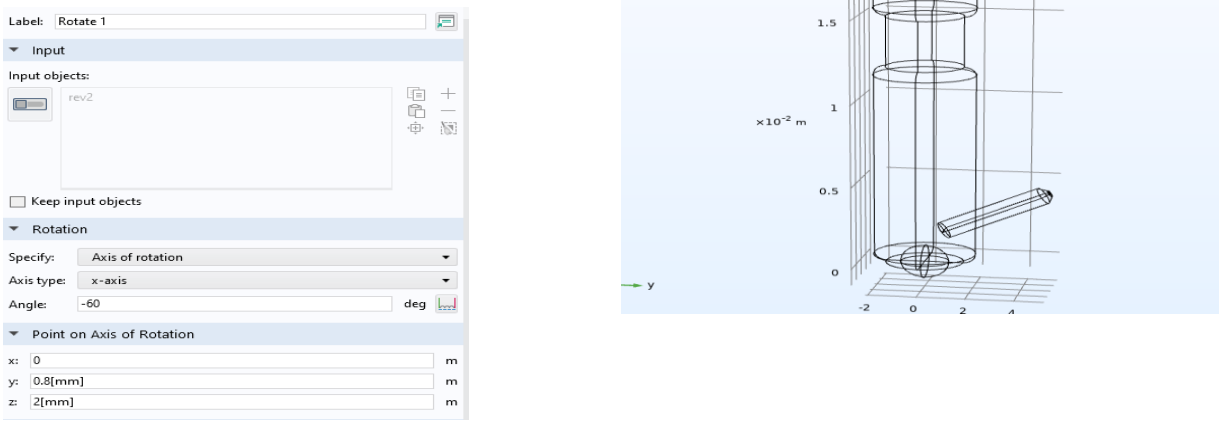


Figure 15: Rotation of the anchoring arm

11. To generate 4 anchoring arms, the 1st anchoring arm had been rotated with range (90, 360/4, 360).

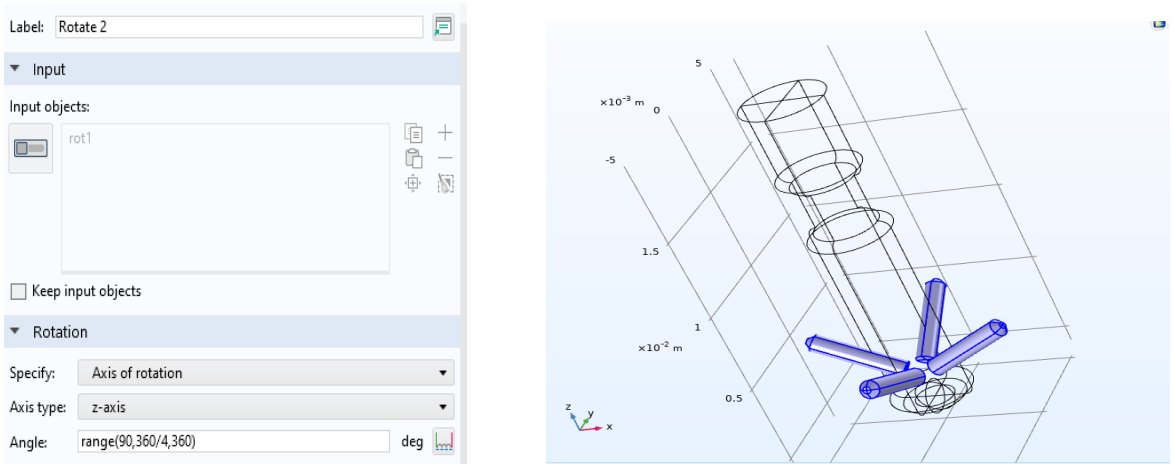
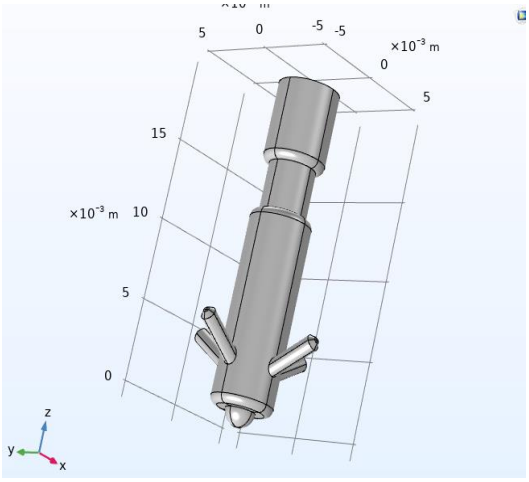


Figure 16: The 4 anchoring arms after 2nd

12. Then the final geometry (3D model) has been generated after forming union [7][8].



Label: Parameters 1

Parameters

Name	Expression	Value	Description
rect1_w	2.1[mm]	0.0021 m	
rect1_h	20[mm]	0.02 m	
rect2_w	0.5[mm]	5E-4 m	
rect2_h	3[mm]	0.003 m	
fil_rad	0.5[mm]	5E-4 m	
sph_electr...	1[mm]	0.001 m	
wp2_rect1...	0.5[mm]	5E-4 m	
wp2_rect1...	5[mm]	0.005 m	
cyl_r	10[mm]	0.01 m	
cyl_h	40[mm]	0.04 m	

Figure 17: Final geometry after forming union and the parameters

4.2 Analysis

1. At first physics (AC/DC>Electric Fields and Currents>Electric Currents (ec)) had been selected from the select physics tree in a separate file under Model wizard. Then, General Studies>Stationary had been selected from the Select Study tree.

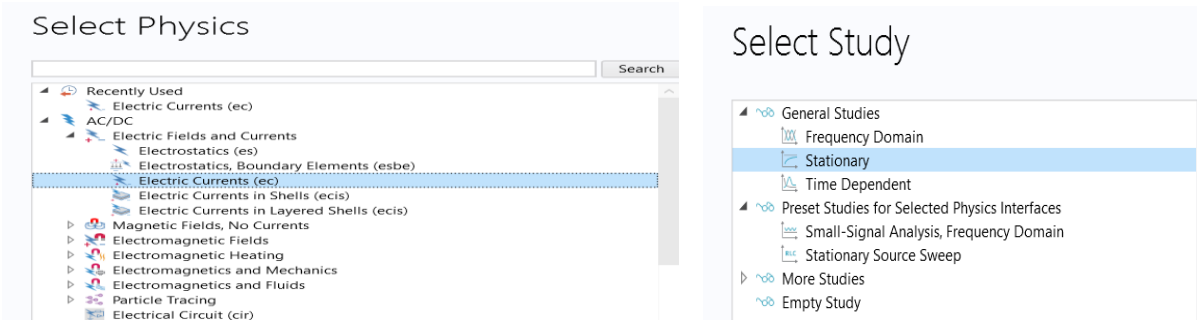


Figure 18: Selection of physics

2. Then the geometry sequence had been inserted from the pacemaker_ electrode_ geom_sequence.mph file.

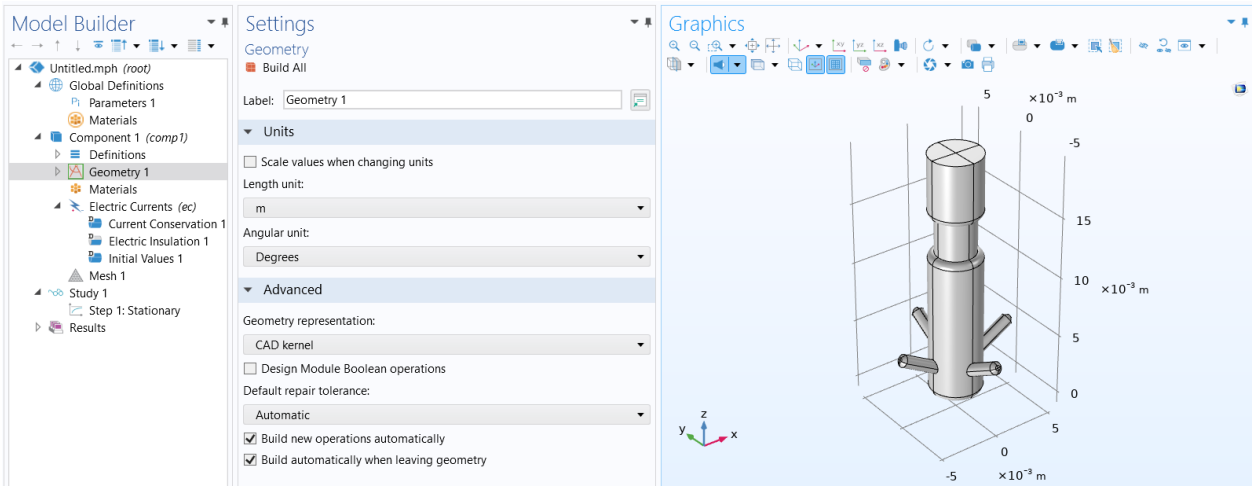


Figure 19: Insertion of the geometry sequence

3. Then the volume surrounding the electrode had been defined as heart tissue. The simulation only takes place in this volume, where the boundaries of the electrode influence the result. The radius of the boundary is 10 mm and height is 40 mm.

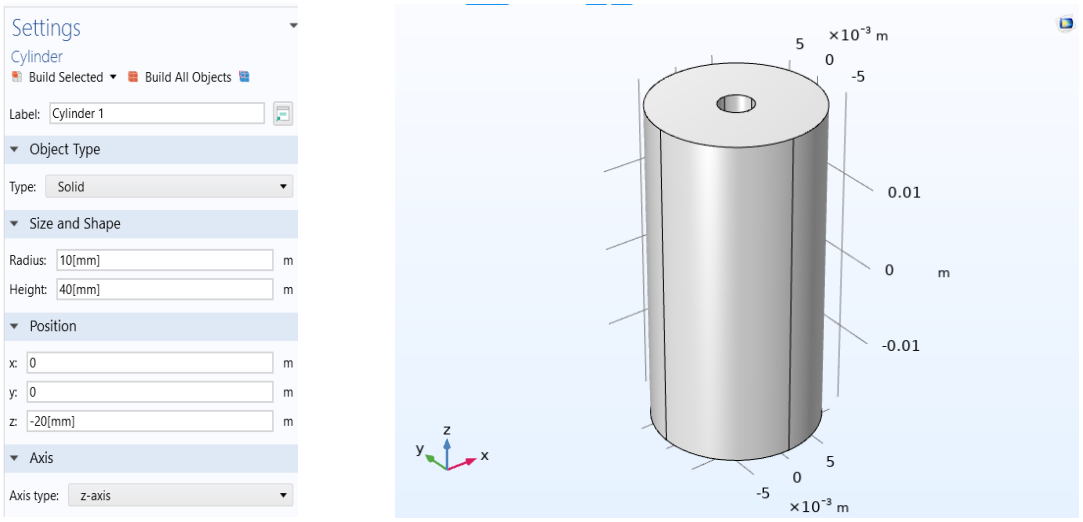


Figure 20: The volume surrounding the electrode defined as heart tissue

4. After that a selection had been defined corresponding to the grounded electrode (counter electrode) for later use. On the object fin, boundaries 29, 30, 58, and 63 had been selected only to be defined as the counter electrode.

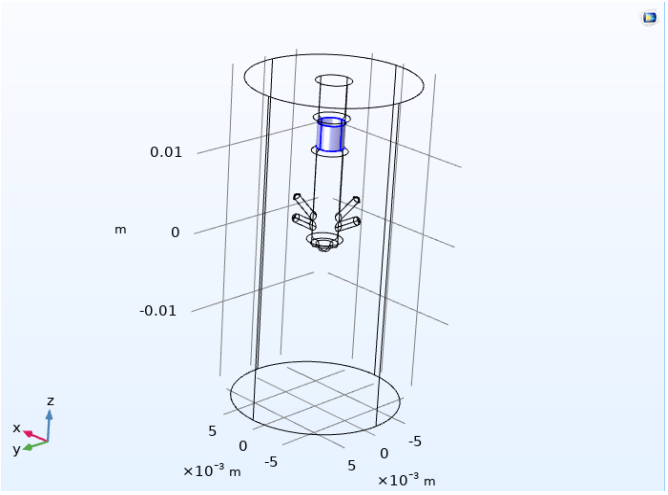


Figure 21: Selection of the counter electrode (grounded electrode)

5. A blank material had been defined to represent the heart tissue since the electrode is inserted into the human heart. So, the electrical conductivity (0.4 S/m) and relative permittivity had been selected for the heart tissue [7][8].

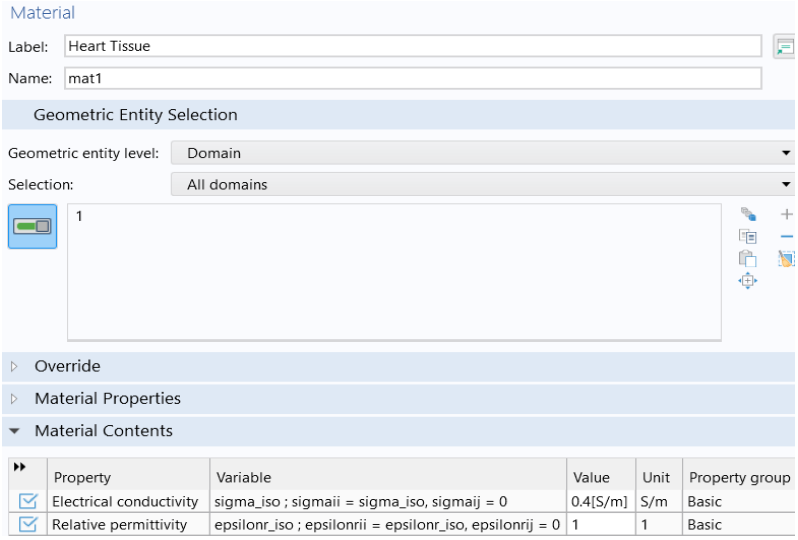


Figure 22: Selection of the electrical conductivity (0.4 S/m) and relative permittivity for the heart tissue

6. Selection of insulating material Silicone Carbide, PTFE (Polytetrafluoroethylene) for the body

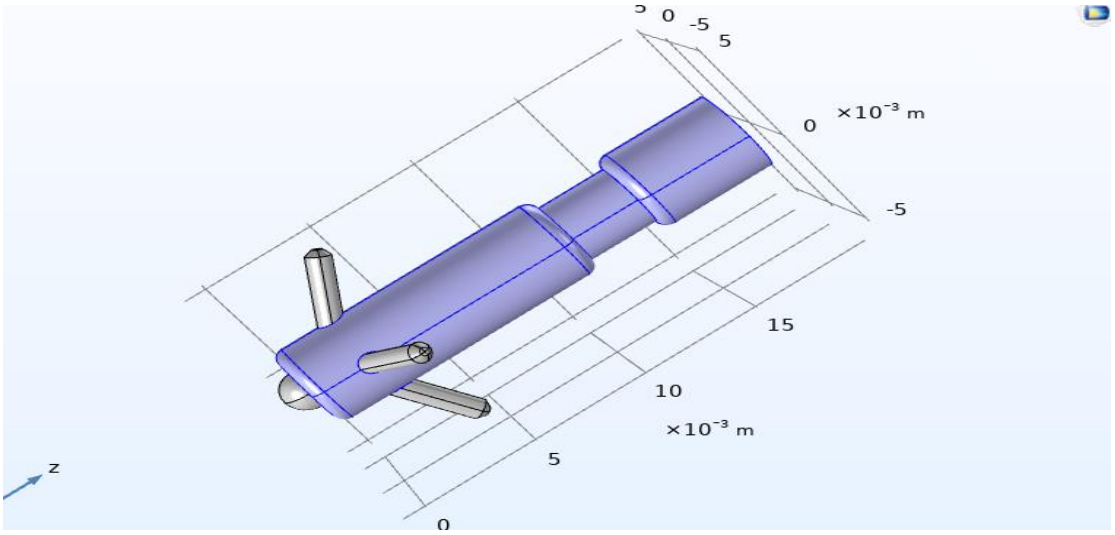


Figure 23: Selection of Silicone Carbide and for PTFE(Polytetrafluoroethylene) the electrode body

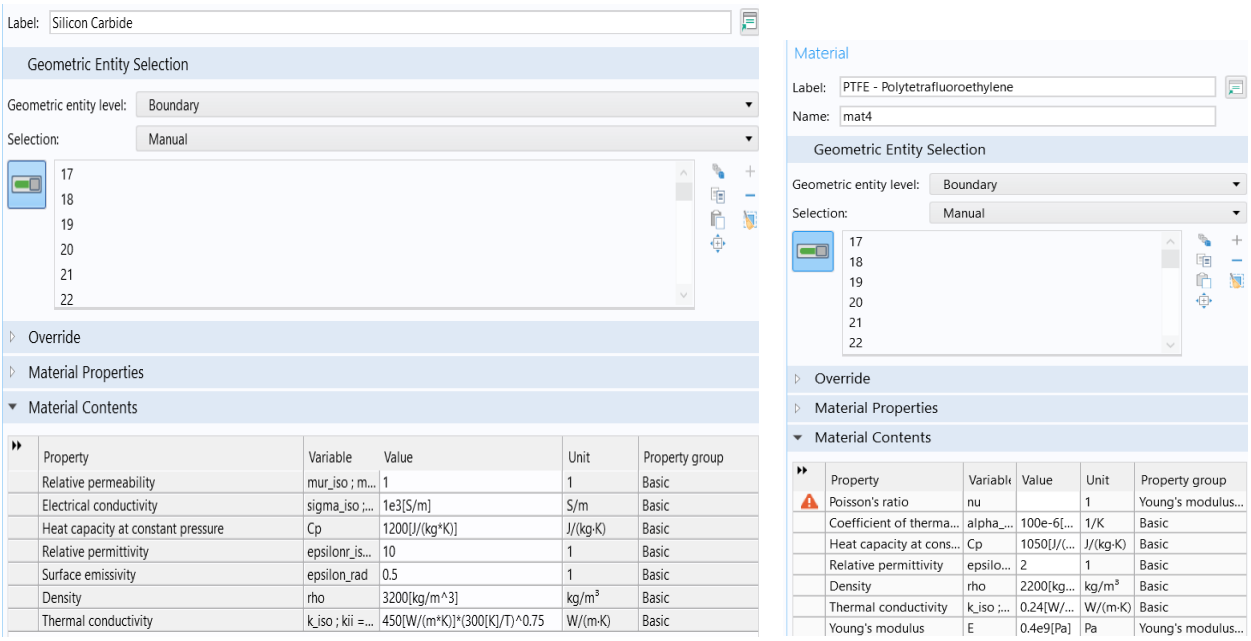


Figure 24: Properties of the material Silicone Carbide and PTFE(Polytetrafluoroethylene)

7. Selection of PIB (polyisobutylene or butyl rubber) [solid] for the tines.

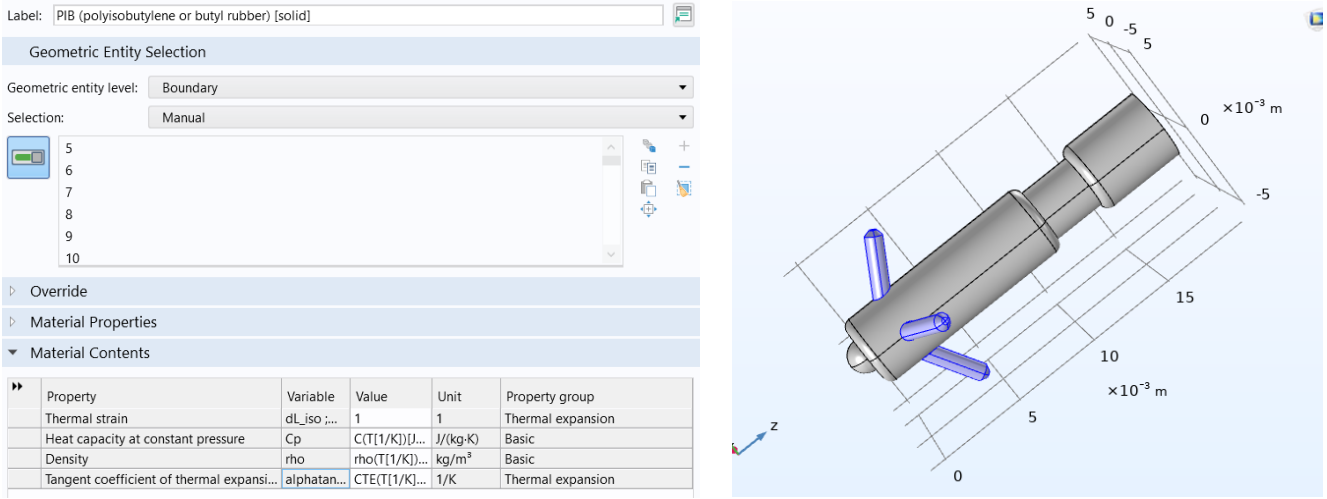


Figure 25: Selection of rubber for the tines and the properties of the material

8. Selection of conducting material $CoCr_2O_4$, Platinum, and Titanium alloy (Ti Grade 1 (solid, annealed)) for the spherical electrode [9][10][11].

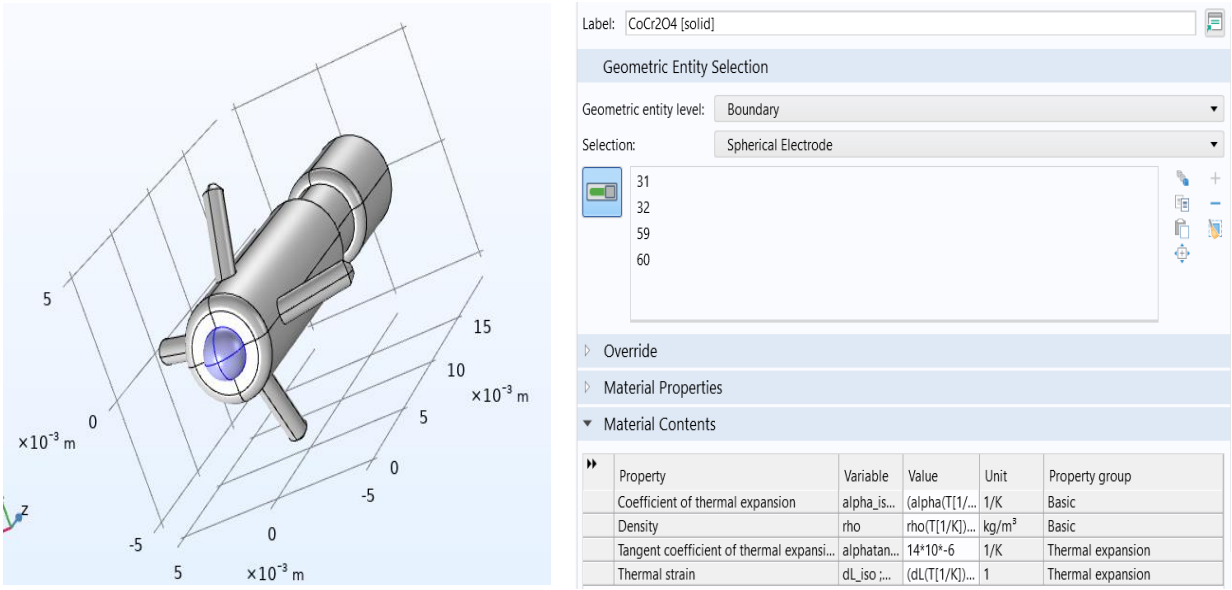
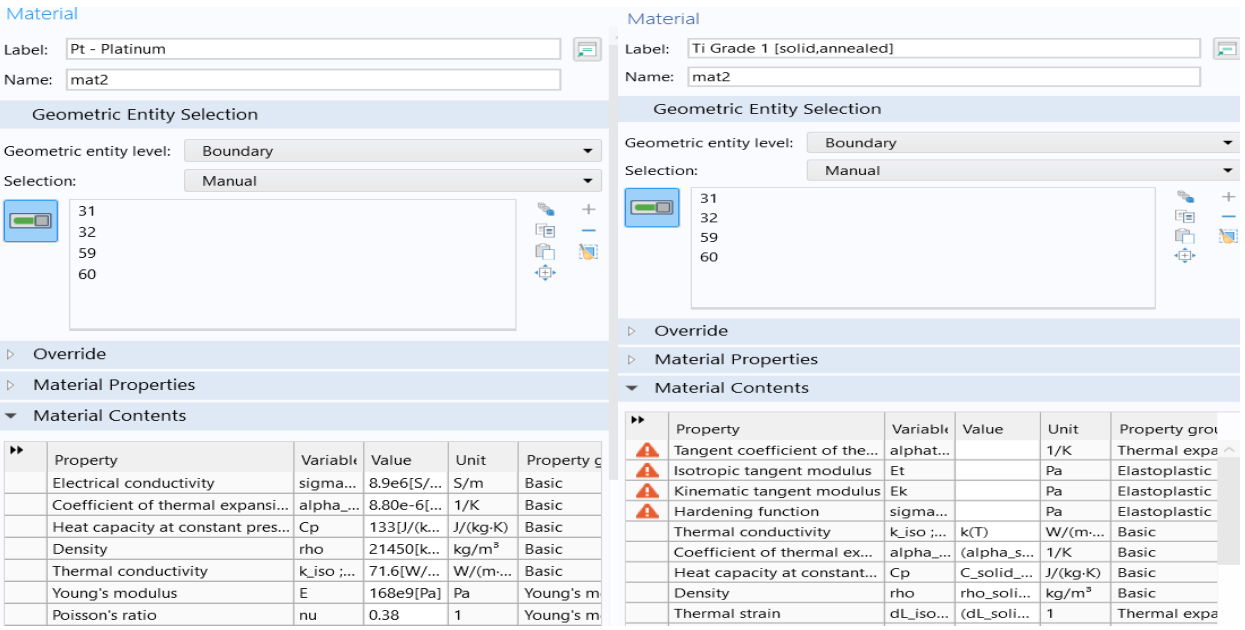


Figure 26: Selection of $CoCr_2O_4$, Platinum, and Titanium alloy (Ti Grade 1 (solid, annealed)) for the spherical electrode and the properties of the material

a)Properties of $CoCr_2O_4$



b)Properties of Platinum c)Properties of Titanium alloy (Ti Grade 1 (solid, annealed))

9. i) Then counter electrode (boundaries 29, 30, 58, and 63) had been selected for ground from the Electric current(ec) section of component 1.

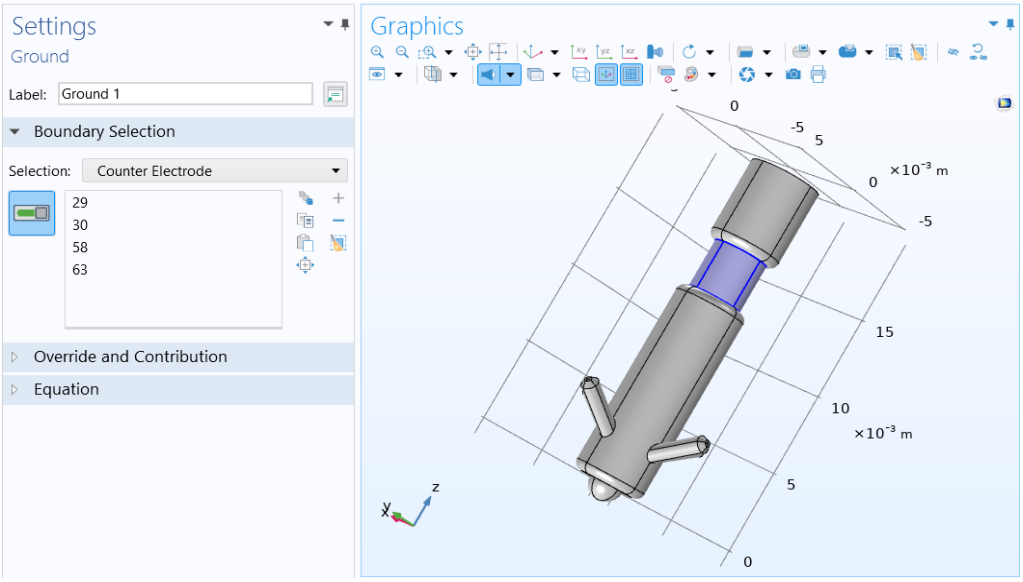


Figure 27: Selection of the ground

ii) For applying electric potential, the spherical (boundaries 31, 32, 59, and 60) electrode had been selected and 1 V electric potential had been applied [7][8].

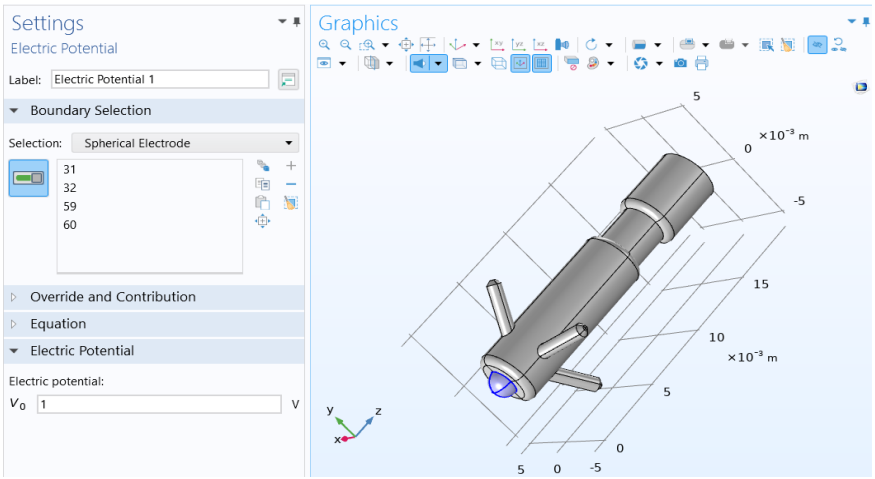
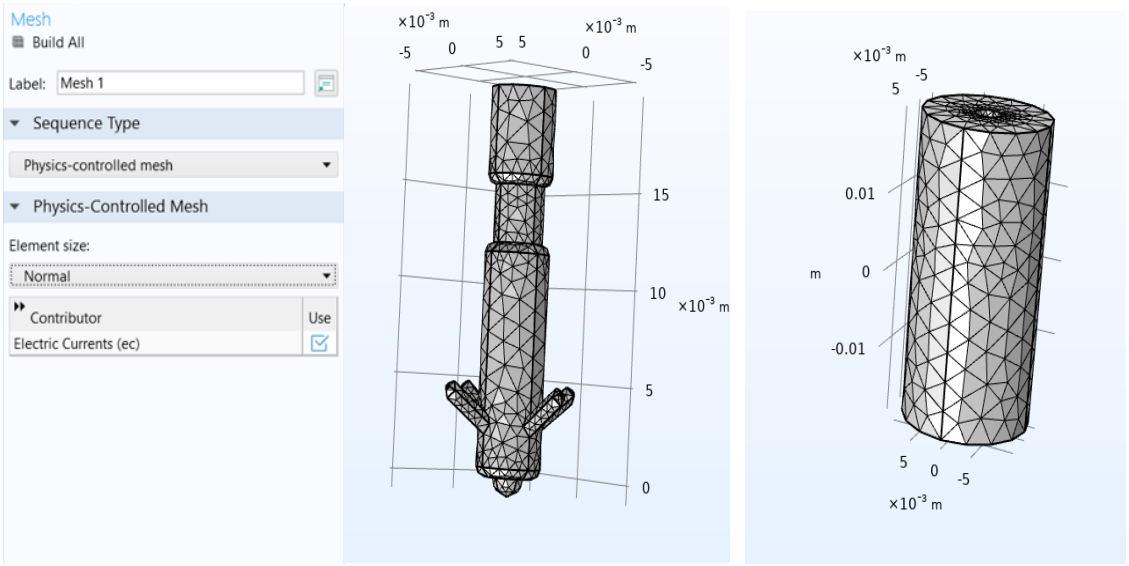


Figure 28: Selection of the spherical working electrode and application of 1V electric potential

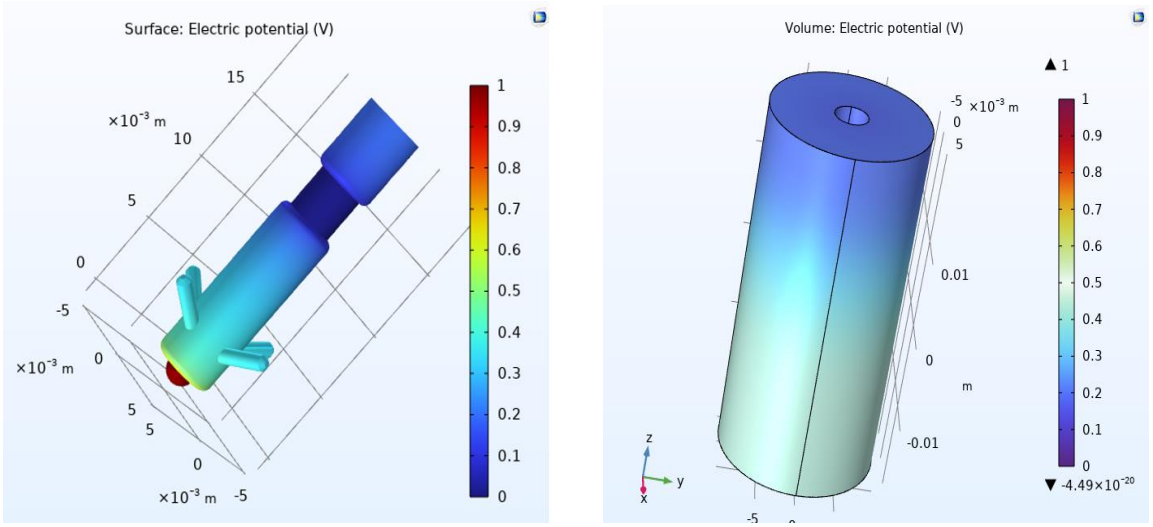
10. Then mesh has been applied where the element size was normal.



(a) (b) Boundary (heart tissue)

Figure 29: The default physics-controlled mesh (element size

11. Then for different materials the electric potential distribution had been computed from the electric potential (ec) section by using the default settings for the stationary solver where the radius of spherical electrode(1 mm) is fixed.



(a) Electrode (b) Boundary

Figure 30: The electrostatic potential distribution on the surface of electrode and boundary where $CoCr_2O_4$ is used for the spherical electrode , PIB (polyisobutylene or butyl rubber(solid))is used for the tines ,and Silicone Carbide for the electrode body

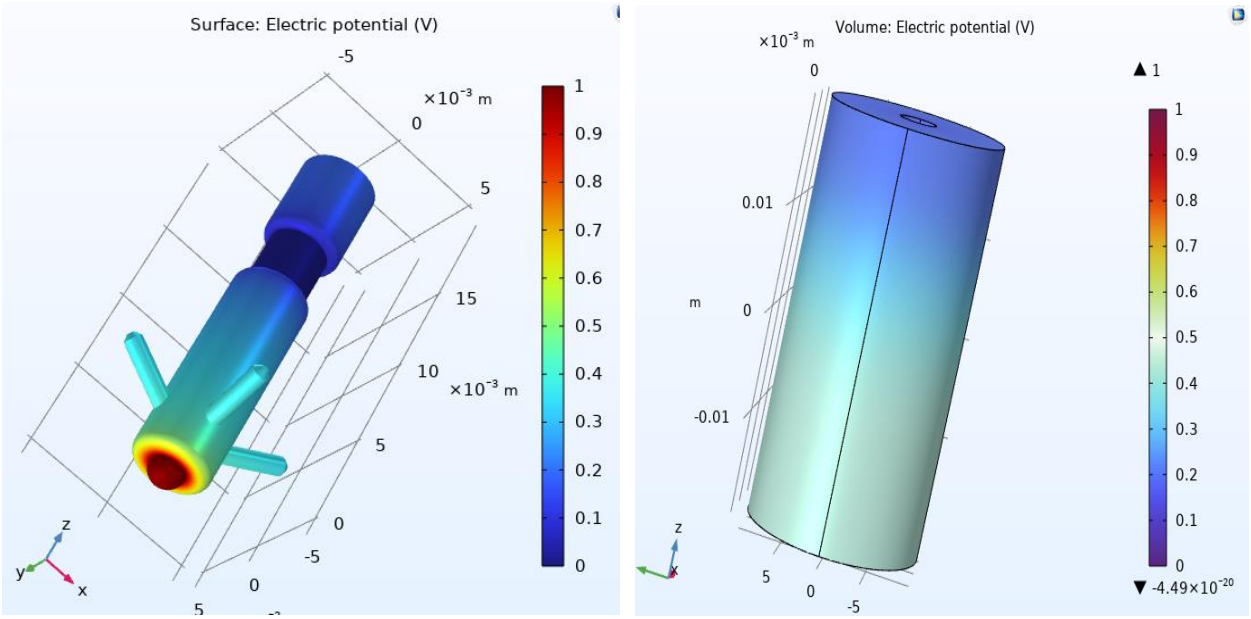


Figure 31: The electrostatic potential distribution on the surface of electrode and boundary where Platinum is used for the spherical electrode , PIB (polyisobutylene or butyl rubber(solid)) is used for the tines ,and PTFE (Polytertafluoroethylene) for the electrode body

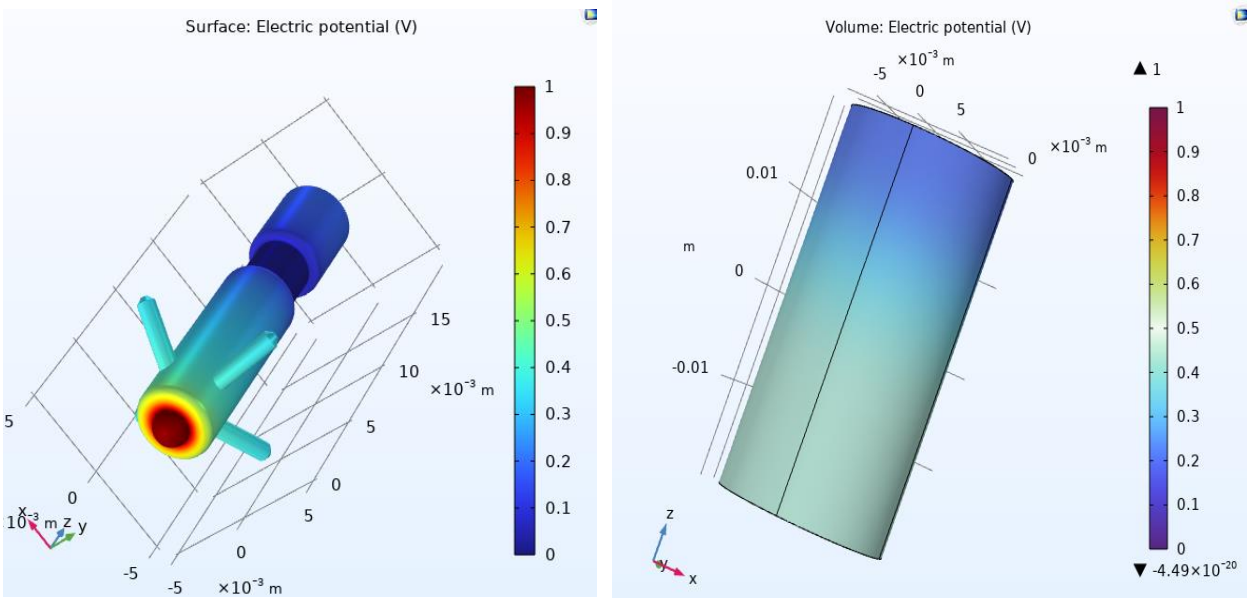


Figure 32: The electrostatic potential distribution on the surface of electrode and boundary where Titanium alloy (Ti Grade 1 (solid, annealed)) is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines ,and PTFE (Polytertafluroethylene) for the electrode body

12. Then for different material the surface plot of the electric potential had been combined with streamlines for visualizing the total current density where the radius of the spherical electrode (1 mm) is fixed . Where the radius of the tube of the streamline was (min(ec.normJ/0.1[mA/mm^2],1)* 0.2[mm]). ec.normJ is the variable for the current density norm. This expression states that tubes are 0.2[mm] wide in the points having 0.1[mA/mm^2]. The "min" function saturates the increase of the tube radius where the current density approaches 0.1[mA/ mm^2] [7][8].

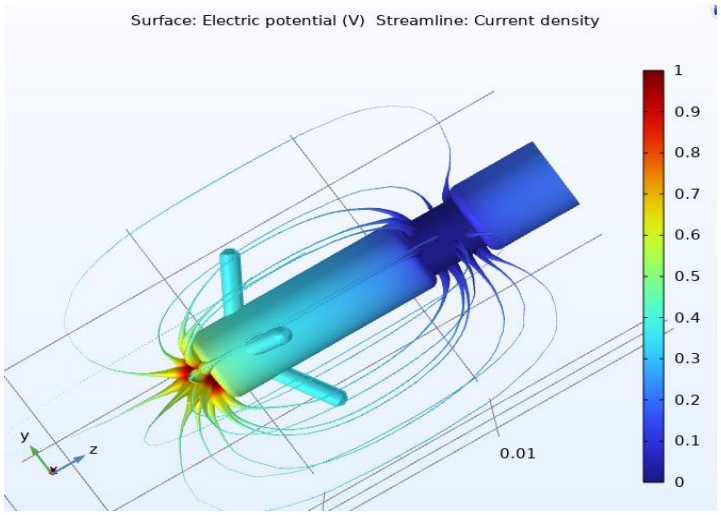


Figure 33: The total current density is shown as streamlines and the electrostatic potential distributed on the surface of the electrode where $CoCr_2O_4$ is used for the spherical electrode , PIB (polyisobutylene or butyl rubber(solid))is used for the tines ,and Silicone Carbide for the electrode body

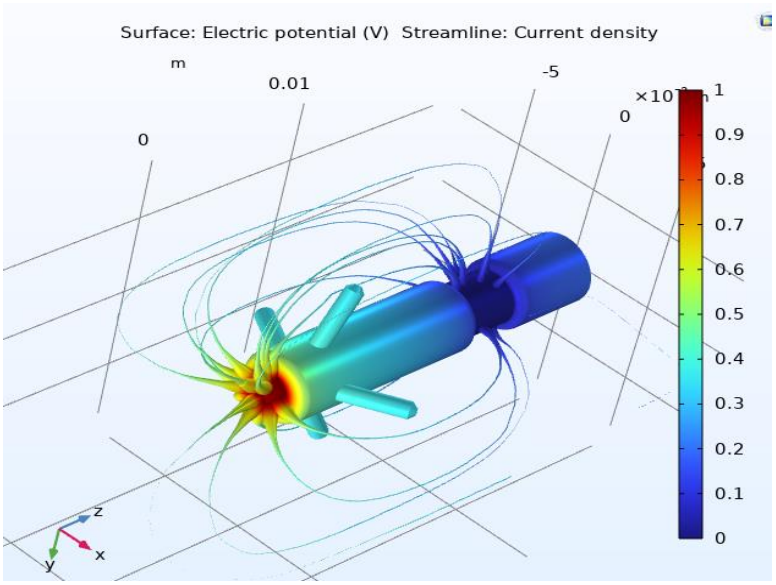


Figure 34: The total current density is shown as streamlines and the electrostatic potential distributed on the surface of the electrode where Platinum is used for the spherical electrode , PIB (polyisobutylene or butyl rubber(solid)) is used for the tines ,and PTFE (Polytertafluoroethylene) for the electrode body

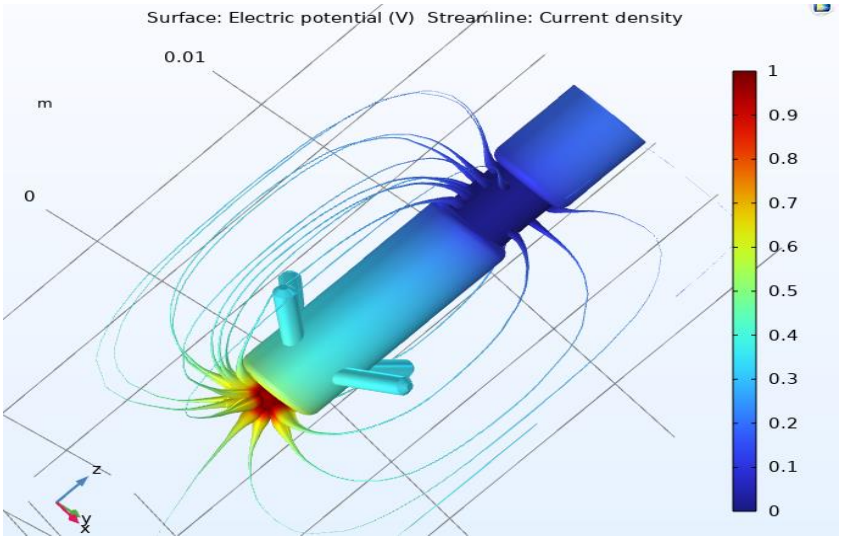


Figure 35: The total current density is shown as streamlines and the electrostatic potential distributed on the surface of the electrode where Titanium alloy (Ti Grade 1 (solid, annealed)) is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines ,and PTFE (Polytertafluroethylene) for the electrode body

13. Then voltage distribution and current distribution had been analyzed for different radius (0.5mm and 1.5 mm) of spherical electrode where Titanium alloy (Ti Grade 1 (solid, annealed)) is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines ,and PTFE (Polytertafluroethylene) for the electrode body.

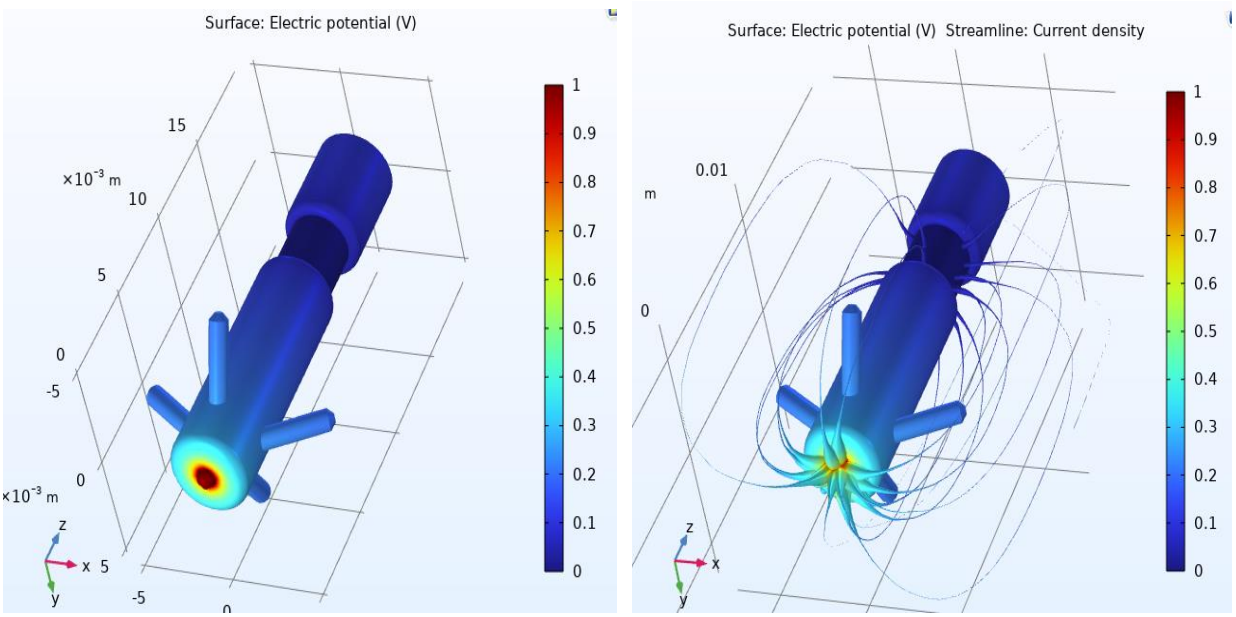


Figure 36: The electrostatic potential distribution on the surface of electrode the total current density is shown as streamlines where the radius of the electrode is 0.5 mm

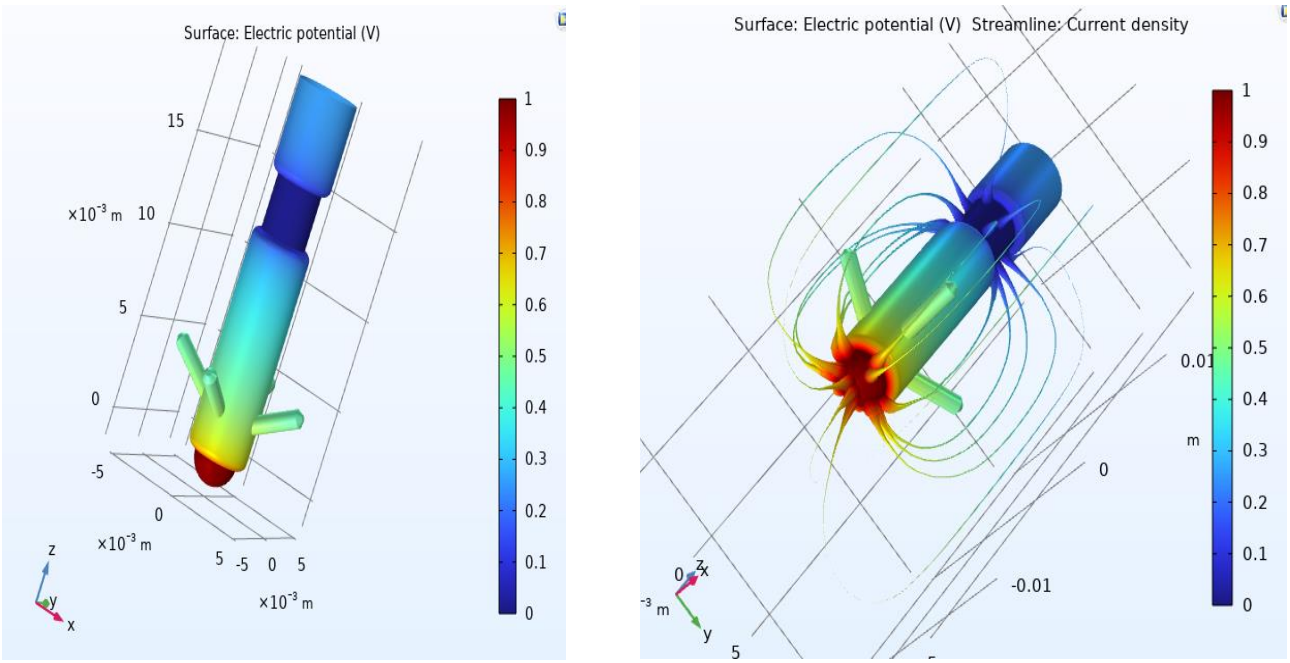


Figure 37: The electrostatic potential distribution on the surface of electrode the total current density is shown as streamlines where the radius of the electrode is 1.5 mm

Chapter 5

Result and Discussion

This simulation gives the potential distribution on the electrode surface and streamlines of the current distribution inside the human heart. As expected, the current density is highest at the small hemisphere, which is the one that causes the excitation of the heart. The current density is fairly uniform on the working electrode. The counter electrode is larger and there are also larger variations in current density on its surface. Mainly, the current is lower with increasing distance from the working electrode. The model shows that the anchoring arms of the device have little influence on the current density distribution[8].

In our experimental setup, various materials were applied to the pacemaker electrode, while no specific material was applied to the surrounding heart tissues. To address this, we delineated the two key properties of heart tissue:

(i) electrical conductivity, measured at 0.4 S/m and

(ii)relative permeability, set at 1.

We chose not to assign a specific material to the heart tissues because their natural properties, such as electrical conductivity and relative permeability, already accurately represented their behavior in the simulation. Also we applied 1V in the electrical potential sector.

Initially, three scenarios were investigated, each involving variations in the materials employed across different components of the pacemaker electrode. These scenarios were delineated as follows:

Case 1: CoCr2O4 is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines, and Silicone Carbide for the electrode body

Case 2: Platinum is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines, and PTFE (Polytetrafluoroethylene) for the electrode body

Case 3: Titanium alloy (Ti Grade 1 (solid, annealed)) is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines, and PTFE (Polytetrafluoroethylene) for the electrode body.

The analysis revealed that, despite the variations in materials used for different parts of the pacemaker electrode, there were minimal differences in both current density and voltage distribution among the scenarios. Regardless of the materials employed, the overall patterns of current density and voltage distribution remained consistent across all scenarios.

The middle layer of the heart muscle is called myocardium and it is responsible for conduction of electric impulses. Therefore, for optimized performance of the pacemaker electrode, the radius of the active electrode should be chosen such that there is maximum current density at the myocardium of the heart. Too much increasing of the tip size would decrease the electric influence on the heart cells whereas, decreasing it too much would cause localized tissue traumas [1].

For comparative analysis, three distinct cases were examined, each involving variations in the radius of the spherical electrode (working electrode). These cases encompassed radius measurements of 0.5 mm, 1 mm, and 1.5 mm, allowing for a comprehensive exploration of the impact of radius size on the electrode's performance and efficacy. The materials we took for this case were Titanium alloy (Ti Grade 1 (solid, annealed)) is used for the spherical electrode, PIB (polyisobutylene or butyl rubber(solid)) is used for the tines, and PTFE (Polytetrafluoroethylene) for the electrode body. From the observations, it was evident that the current density associated with the 0.5 mm radius was insufficient to exert any influence on the heart cells for conduction. Conversely, in the case of the 1.5 mm radius the current density the current density significantly exceeded safe levels, posing a risk of potential damage to the heart muscle. The current density observed with a 1 mm radius exhibited a precise and well-distributed gradient.

Chapter 6

Conclusion & Future scopes

Through the analysis of the pacemaker electrode, it has been determined that an electrode radius of 1 mm gives us the optimum current density. Increasing the voltage between electrodes further enhances current density due to its direct relationship with applied voltage. The spherical electrode was the conducting part. The materials were chosen making sure that the body of the electrode was kept insulated. The design process was done in such a way that the tines were providing proper support for the pacemaker electrode inside the body.

Moreover, our study have opened many doors for further research and experiments.

Future Scope:

1. Investigate the effect of spacing on other electrical parameters such as degrees of freedom and electrical resistance.

2. Explore alternative electrode materials to optimize conductivity, impacting current density.

3. Study the influence of additional parameters, such as electrode width, on current density.

4. Conduct research on the overall size of the pacemaker electrode to refine design considerations.

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